

ClinGen PTEN Expert Panel Specifications to the ACMG/AMP Variant Interpretation Guidelines for PTEN Version 3.2.0

Affiliation: [PTEN VCEP](#)

Type: Richards et.al., 2015 - Combining rules

Description : ACMG Classification Rules Specified for PTEN Variant Curation

Version : 3.2.0

Released : 4/6/2026

Release Notes :

Minor Changes:

1. Added disease name and MOI
2. Updated PS3 and PS4 < and > signs to $\geq$ or $\leq$ if necessary

Rules for PTEN

General Comments: Minor Changes: 1. Correct SpliceAI cutoff for BP4 rule 2. Correct the Rules for Combining Criteria 3. Add BLOSUM matrix, Cleveland Clinic core and Pediatric score tables 4. Added disease name, MOI and added underscores for some greater than symbols

Gene: PTEN (HGNC:9588) [↗](#)

Transcripts:

NM_000314.8

HGNC Name: phosphatase and tensin homolog
Disease:

PTEN hamartoma tumor
syndrome (MONDO:0017623)

[↗](#) **Mode of Inheritance:**

Autosomal dominant
inheritance

Criteria & Strength Specifications

PVS1

Original ACMG Summary

Null variant (nonsense, frameshift, canonical $\pm$ 1 or 2 splice sites, initiation codon, single or multi-exon deletion) in a gene where loss of function (LOF) is a known mechanism of disease.

Caveats:

- Beware of genes where LOF is not a known disease mechanism (e.g. GFAP, MYH7).
- Use caution interpreting LOF variants at the extreme 3' end of a gene.
- Use caution with splice variants that are predicted to lead to exon skipping but leave

the remainder of the protein intact.

- Use caution in the presence of multiple transcripts.

VCEP

PTEN EP Specification: Follow SVI guidance, using PTEN-specific information. Per the PVS1 workflow guidance provided in Tayoun et al. 2018 (PMID 30192042), the following will apply:

1. Nonsense, frameshift variants:
 - PVS1 applies to variants predicted to result in nonsense-mediated decay (NMD); the predicted NMD cutoff for PTEN occurs at c.1121 (p.D375).
 - For nonsense or frameshift variants at the 3' end of the gene NOT predicted to result in nonsense-mediated decay, PVS1 may still be applied if the protein is disrupted at or 5' to c.1121 (NM_000314.6). Please see supplementary information in manuscript for evidence supporting this cutoff.
 - PVS1_Moderate applies to variants resulting in protein truncation 3' of this cutoff.
2. Splicing variants (+/- 1,2 intronic positions):
 - Only apply to the variants resulting NMD (please refer to decision tree) OR entire exon deletion:
 - Exons 1,2,4,5,6 OR 7 deletions OR multi-exon deletion: PVS1 (Resulting frameshift)
 - Exons 3,8 OR 9 deletions: PVS1_Strong (in-frame but truncated/alterd region is critical to protein function).
3. Deletion (Single/multi exon to full gene): Please refer to decision tree.
4. Duplication: Please refer to decision tree.
5. Initiation codon: PVS1 applies to initiation codon variants.

PTEN EP Commentary: No known alternative start codon in other transcripts. There are sufficient patients' data from literature and labs support the pathogenicity of initiation codon variants.

Very Strong

Use PTEN PVS1 decision tree.

Modification Disease-specific
Type:

Strong

Use PTEN PVS1 decision tree.

Modification Disease-specific
Type:

Moderate

Use PTEN PVS1 decision tree.

Modification Disease-specific
Type:

PS1

Original ACMG Summary

Same amino acid change as a previously established pathogenic variant regardless of nucleotide change.

Example: Val->Leu caused by either G>C or G>T in the same codon.

Caveat: Beware of changes that impact splicing rather than at the amino acid/protein level.

VCEP **PTEN EP Specification:** PS1 will be applied as described and expanded **Specifications** to include a different nucleotide substitution for an intronic splice site variant if the predicted impact is equal to or greater than the known pathogenic variant per *in silico* splicing tools. Caution should be used when applying this criteria to exonic variants causing aberrant splicing.

Strong

Same amino acid change as a previously established pathogenic variant regardless of nucleotide change OR different variant at same nucleotide position as a pathogenic splicing variant, where in silico models predict impact equal to or greater than the known pathogenic variant.

Modification Disease-specific
Type:

PS2

Original ACMG Summary

De novo (both maternity and paternity confirmed) in a patient with the disease and no family history.

Note: Confirmation of paternity only is insufficient. Egg donation, surrogate motherhood, errors in embryo transfer, etc. can contribute to non-maternity.

VCEP PS2_Very Strong: Two or more occurrences of PS2 OR two or more **Specifications** occurrences of PM6 AND one occurrence of PS2.

Very Strong

Two proven OR four assumed OR one proven + two assumed de novo observations in a patient with the disease and no family history.

Modification Strength

Type:

Strong

De novo (both maternity and paternity confirmed) observation in a patient with the disease and no family history.

Modification None

Type:

PS3

Original ACMG

Summary

Well-established in vitro or in vivo functional studies supportive of a damaging effect on the gene or gene product.

Note: Functional studies that have been validated and shown to be reproducible and robust in a clinical diagnostic laboratory setting are considered the most well-established.

VCEP

PTEN EP Specification:

Specifications:

PS3 may be applied to the following assays:

- RNA, mini-gene, or other assay demonstrating an impact on splicing.

PS3_Moderate:

- Mighell et al. 2018 (PMID: 29706350): Massively parallel functional assay interrogating phosphatase activity.
 - In the supplementary material (Table S2) search for the variant in columns A or B and make sure the variant in question is listed as TRUE under column I (high confidence). If not, do not use as evidence.
 - Under column G, the cumulative score is listed. Apply PS3_moderate for all variants with scores ≤ -1.11 .

PS3_Supporting: Other studies demonstrating lipid phosphatase activity $<50\%$ of wild-type or abnormal *in vitro* cellular assay or transgenic model with phenotype different from wild-type that does not meet

PS3_moderate. Examples of *in vitro* cellular assays to be considered for PS3_supporting evidence may include:

- *In vitro* assay demonstrating $>50\%$ reduction in phosphatase activity compared to wild type control. Phosphatase assays for which criteria may be applied must include a catalytic dead control, such as p.C124S, as well as at least three biological replicates: Myers et al. 1998 (PMID: 9811831), Stambolic et al. 1998 (PMID: 9778245), Han et al. 2000 (PMID: 10866302), Rodriguez-Escudero et al. 2011 (PMID: 21828076), Costa et al. 2015 (PMID: 26504226), Malek et al. 2017 (PMID: 29056325).

- Decreased PTEN or increased pAKT expression: Tan 2011 (PMID: 21194675), Spinelli 2015 (PMID: 25527629).
- Disruption of protein cellular localization: Lobo et al. 2009 (PMID: 19457929), He et al. 2012 (PMID: 22962422), Gil et al. 2015 (PMID: 25875300)
- Aberrant cellular phenotypes, including defective cell migration, proliferation, and invasion: Costa et al. 2015 (PMID: 26504226), Malek et al. 2017 (PMID: 29056325)

Strong

Well-established *in vitro* or *in vivo* functional studies supportive of a damaging effect on the gene or gene product.

- RNA, mini-gene, or other assay shows impact on splicing

Modification Disease-specific

Type:

Moderate

Well-established *in vitro* or *in vivo* functional studies supportive of a damaging effect on the gene or gene product.

- Phosphatase activity ≤ -1.11 per Mighell et al. 2018, PMID: 29706350.

Modification Disease-specific

Type:

Supporting

Phosphatase activity $<50\%$ of wild-type or abnormal *in vitro* cellular assay or transgenic model with phenotype different from wild type that does not meet PS3_moderate.

Modification Disease-specific, Strength

Type:

PS4

Original ACMG Summary

The prevalence of the variant in affected individuals is significantly increased compared to the prevalence in controls.

Note 1: Relative risk (RR) or odds ratio (OR), as obtained from case-control studies, is >5.0 and the confidence interval around the estimate of RR or OR does not include 1.0. See manuscript for detailed guidance.

Note 2: In instances of very rare variants where case-control studies may not reach statistical significance, the prior observation of the variant in multiple unrelated patients with the same phenotype, and its absence in controls, may be used as moderate level of evidence.

VCEP

Specifications **PTEN EP Commentary:** This criterion is unlikely to be used in this manner for a condition as rare as PHTS. However, if sufficiently powered, a case-control study finding an odds ratio >2 for a PHTS component phenotype with $p < 0.05$ and 95% confidence interval with lower limit >1.5 , this criteria may be applied. However, this criterion may *not* be applied in combination with PP4.

- Use 2: Patient's phenotype or family history is highly specific for a disease with a single genetic etiology.

PTEN EP Specifications: This criterion may not be applied if BS1 applies. Phenotype specificity scores are added across independent probands and calculated as follows:

Adults:

- 1 point per proband with Cleveland Clinic (CC) score ≥ 30 (Tan 2011)
- 0.5 points per proband with CC score of 25-29.

Children:

- 1 point per proband with pediatric phenotype score ≥ 5 (please see supplementary information in manuscript for scoring rubric).
- 0.5 points per proband with pediatric phenotype score of 4, but autism/developmental delay/intellectual disability may not contribute to the score.

PS4_Very Strong: Probands with specificity score ≥ 16 .

PS4: Probands with specificity score of 4-15.5.

PS4_Moderate: Probands with specificity score of 2-3.5.

PS4_Supporting: Proband(s) with specificity score of 1-1.5.

Very Strong

Probands with specificity score ≥ 16 (see text).

Modification Strength

Type:

Strong

Probands with specificity score 4-15.5 (see text) OR The prevalence of the variant in affected individuals is significantly increased compared with the prevalence in controls.

Modification Strength

Type:

Moderate

Probands with specificity score of 2-3.5 (see text).

Modification Strength

Type:

Supporting

Phenotype specific for disease with single genetic etiology. Proband(s) with specificity score of 1-1.5 (see text).

Modification Disease-specific

Type:

PM1

Original ACMG

Summary

Located in a mutational hot spot and/or critical and well-established functional domain (e.g. active site of an enzyme) without benign variation.

VCEP **PTEN EP Specification:** Defined to include residues in one of PTEN's catalytic motifs, which include the WPD loop (residues 90-94), P-loop (also described as phosphatase core, residues 123-130), and the TI-loop (residues 166-168) (NP_000305.3) (Lee 1999).

Moderate

Located in a mutational hot spot and/or critical and well-established functional domain. Defined to include residues in catalytic motifs: 90-94, 123-130, 166-168 (NP_000305.3)

Modification Disease-specific

Type:

PM2

Original ACMG

Summary

Absent from controls (or at extremely low frequency if recessive) in Exome Sequencing Project, 1000 Genomes or Exome Aggregation Consortium.
Caveat: Population data for indels may be poorly called by next generation sequencing.

VCEP **PTEN EP Specification:** Criteria may be applied if a variant is present at ≤ 0.00001 (0.001%) allele frequency in gnomAD or another large sequenced population. If multiple alleles are present within a subpopulation, allele frequency in that subpopulation must be < 0.00002 (0.002%). Please see supplementary information in manuscript supporting application of PM2 for ultra-rare alleles.

Supporting

Absent in population

- Databases present at <0.00001 (0.001%) allele frequency in gnomAD or another large sequenced population. If multiple alleles are present within any subpopulation, allele frequency in that subpopulation must be <0.00002 (0.002%).

Modification Disease-specific
Type:

PM3

Original ACMG Summary

For recessive disorders, detected in trans with a pathogenic variant
Note: This requires testing of parents (or offspring) to determine phase.

Not Applicable

Comments: This rule is not applicable to PTEN.

PM4

Original ACMG Summary

Protein length changes due to in-frame deletions/insertions in a non-repeat region or stop-loss variants.

VCEP **PTEN EP Specification:** For in-frame insertions or deletions, criteria may apply only if the variant impacts at least one residue in one of the catalytic motifs specified in the PM1 criteria. Criteria will also apply for variants resulting in protein extension.

Moderate

Protein length changes due to in-frame deletions/insertions in a non-repeat region or stop-loss variants. Applies to in-frame insertions or deletions impacting at least one residue in a catalytic motif (see PM1), and variants causing protein extension.

Modification Disease-specific
Type:

PM5

Original ACMG Summary

Novel missense change at an amino acid residue where a different missense change determined to be pathogenic has been seen before.
Example: Arg156His is pathogenic; now you observe Arg156Cys.

Caveat: Beware of changes that impact splicing rather than at the amino acid/protein level.

VCEP **PTEN EP Specifications:**

Specifications:

- This rule may be applied when the known variant is likely pathogenic unless applying would lead to a higher (pathogenic) classification for the variant being assessed.
- The variant in question need not be novel but must have a BLOSUM62 (Henikoff 1992) score equal to or less than the known variant.

Moderate

Missense change at an amino acid residue where a different missense change determined to be pathogenic or likely pathogenic has been seen before. In addition, variant being interrogated must have BLOSUM62 score equal to or less than the known variant.

Modification Disease-specific

Type:

PM6

Original ACMG Summary

Assumed de novo, but without confirmation of paternity and maternity.

VCEP PM6_Very Strong: Four or more occurrences of PM6 OR two occurrences of
Specifications PM6 AND one occurrence of PS2.

PM6_Strong: Two occurrences of PM6 OR occurrence of PM6 for an individual with a highly specific phenotype (meets criteria to count towards PS4).

- Of note, when PM6_S is applied for a single individual with phenotype specificity, the individual will not be counted towards PS4 as well.

Very Strong

Two proven OR four assumed OR one proven + two assumed de novo observations in a patient with the disease and no family history.

Modification Strength

Type:

Strong

Two probands with presumed *de novo* occurrence (maternity/ paternity not confirmed) with the disease and no family history.

- May also be used for a proband with presumed *de novo* occurrence for an individual with a highly specific phenotype (meets criteria to count towards PS4)

Modification Strength

Type:

Moderate

Assumed *de novo*, but without confirmation of paternity and maternity, in proband with the disease and no family history.

Modification None

Type:

PP1

Original ACMG

Summary

Co-segregation with disease in multiple affected family members in a gene definitively known to cause the disease.

Note: May be used as stronger evidence with increasing segregation data.

VCEP PTEN EP Specification:

Specifications:

PP1: Requires 3 or 4 meioses in order to apply.

PP1_Strong: At least 7 meioses required across at least two families.

PP1_Moderate: Requires 5 or 6 meioses in order to apply.

Strong

Co-segregation with disease in multiple affected family members, with ≥ 7 meioses observed across at least two families.

Modification Strength

Type:

Moderate

Co-segregation with disease in multiple affected family members, with 5 or 6 meioses observed.

Modification Strength

Type:

Supporting

Co-segregation with disease in multiple affected family members, with 3 or 4 meioses observed.

Modification Disease-specific

Type:

PP2

Original ACMG Summary

Missense variant in a gene that has a low rate of benign missense variation and where missense variants are a common mechanism of disease.

Supporting

Missense variant in a gene that has a low rate of benign missense variation and where missense variants are a common mechanism of disease.

Modification None

Type:

PP3

Original ACMG Summary

Multiple lines of computational evidence support a deleterious effect on the gene or gene product (conservation, evolutionary, splicing impact, etc.).

Caveat: As many *in silico* algorithms use the same or very similar input for their predictions, each algorithm should not be counted as an independent criterion. PP3 can be used only once in any evaluation of a variant.

VCEP

PTEN EP Specification: To be applied to synonymous or intronic variants where SpliceAI and VarSeak *in silico* models predict a splicing impact (SpliceAI: scores 0.5-1 are consider evidence of pathogenic. VarSeak: Class 4 and 5 are consider evidence of pathogenic). May also be applied to missense variants with REVEL score > 0.7.

PTEN EP Commentary: Per Bayesian adaptation of the ACMG/AMP variant interpretation framework (Tavtigian et al., 2018), odds of pathogenicity (OddsPath) were estimated for various numbers of previously classified controls. When REVEL scores > 0.7 were used as evidence of pathogenic and < 0.5 were used as evidence of benign, the oddsPath was equated with moderate evidence strength for pathogenic conditions. Given that the VCEP also applies PP2 for missense variants, we decided to downgrade the evidence strength to be used at a supporting level.

Supporting

Multiple lines of computational evidence support a deleterious effect on the gene or gene product.

- Splicing variants: Concordance of SpliceAI and VarSeak

- Missense variants: REVEL score > 0.7

Modification Disease-specific
Type:

PP4

**Original ACMG
Summary**

Patient’s phenotype or family history is highly specific for a disease with a single genetic etiology.

Not Applicable

Comments: PTEN EP Commentary: Phenotype specificity has been incorporated into the rule specifications for PS4 Use 2.

PP5

**Original ACMG
Summary**

Reputable source recently reports variant as pathogenic, but the evidence is not available to the laboratory to perform an independent evaluation.

Not Applicable

This criterion is not for use as recommended by the ClinGen Sequence Variant Interpretation VCEP Review Committee. [PubMed : 29543229](#) 

BA1

**Original ACMG
Summary**

Allele frequency is above 5% in Exome Sequencing Project, 1000 Genomes or Exome Aggregation Consortium.

VCEP Specifications To be applied for variants with filtering allele frequency >0.00056 (>0.056%) in gnomAD. Please see information in BS1 section for data supporting this cutoff.

Stand Alone
gnomAD Filtering allele frequency >0.00056 (0.056%)

Modification Disease-specific
Type:

BS1

Original ACMG

Summary

Allele frequency is greater than expected for disorder.

VCEP

PTEN EP Specification:

Specifications:

BS1: To be applied for variants with filtering allele frequency of 0.000043 up to 0.00056 (0.0043% up to 0.056%) in gnomAD.

BS1_Supporting: To be applied for variants with filtering allele frequency of 0.0000043 up to 0.000043 (0.00043% up to 0.0043%) in gnomAD.

BA1, BS1, and BS1_P thresholds are based on the approach published by Whiffin et al. (PMID 28518168) using the following values:

- Prevalence: 1 in 9,000 (based on 15 disease-associated alleles present among the gnomAD population of ~135,000 individuals)
- Allelic heterogeneity: 22/282 (based on prevalence of most common pathogenic *PTEN* variants, p.R130X and p.R335X, per Tan et al. PMID 21194675 and Bubien 2013 PMID 23335809)
- Penetrance: 10% (overall cancer by age 40 for men with pathogenic germline *PTEN* variants is approximately 20% per Bubien 2013 PMID 23335809)

Using these data points results in a BS1 value of 0.000043. BA1 was calculated by setting allelic heterogeneity to 1, and BS1_P by reducing BS1 by an order of magnitude.

Strong

gnomAD Filtering allele frequency from 0.000043 (0.0043%) up to 0.00056 (0.056%)

Modification Disease-specific

Type:

Supporting

Allele frequency from 0.0000043 (0.00043%) up to 0.000043 (0.0043%).

Modification Disease-specific, Strength

Type:

BS2

Original ACMG

Summary

Observed in a healthy adult individual for a recessive (homozygous), dominant (heterozygous), or X-linked (hemizygous) disorder, with full penetrance expected at an early age.

VCEP**PTEN EP Specifications:****Specifications:**

BS2: Variant must be observed in the homozygous state in a healthy or PHTS-unaffected individual. Two independent observations are required if the homozygous status is not confirmed via parental testing. If BS1 is also applied, this criteria will be applied at the supporting evidence level to avoid a variant reaching benign status solely based on homozygous occurrences due to high population frequency (BS1+BS2).

BS2_Supporting: Two homozygous observations with no clinical data provided, or meets criteria for BS2 but BS1 is also applied.

Strong

Observed in the homozygous state in a healthy or PHTS-unaffected individual. One observation if homozygous status confirmed, two if not confirmed. To be applied at supporting evidence level if BS1 is also applied.

Modification Disease-specific

Type:**Supporting**

Two homozygous observations with no clinical data provided, or meets criteria for BS2 but BS1 is also applied.

Modification Disease-specific, Strength

Type:**BS3****Original ACMG
Summary**

Well-established in vitro or in vivo functional studies show no damaging effect on protein function or splicing.

VCEP**PTEN EP Specifications:****Specifications:**

BS3: For intronic or synonymous variants: RNA, mini-gene, or other assay demonstrate no impact on splicing.

BS3_Supporting: *In vitro* or *in vivo* functional study or studies showing no damaging effect on protein function.

PTEN EP Specifications: BS3_supporting may be applied to the following assays:

- Mighell et al. 2018 (PMID: 29706350): Massively parallel functional assay interrogating phosphatase activity.
 - In the supplementary material (Table S2) search for the variant in columns A or B and make sure the variant in question is listed

as TRUE under column I (high confidence). If not, do not use as evidence.

- Under column G, the cumulative score is listed. Apply BS3_supporting for all variants with scores > 0.
- For missense variants: Other studies showing lipid phosphatase activity comparable to wild type in addition to a second assay appropriate to the protein domain demonstrating no statistically significant difference from wild type. Phosphatase assays for which criteria may be applied must include a catalytic dead control, such as p.C124S (NP_000305.3), as well as at least three biological replicates: Myers et al. 1998 (PMID: 9811831), Stambolic et al. 1998 (PMID: 9778245), Han et al. 2000 (PMID: 10866302), Rodriguez-Escudero et al. 2011 (PMID: 21828076), Costa et al. 2015 (PMID: 26504226), Malek et al. 2017 (PMID: 29056325)
- Examples of second assays may include:
 - Decreased PTEN or increased pAKT expression: Tan et al. 2011 (PMID: 21194675), Spinelli et al. 2015 (PMID: 25527629).
 - Disruption of protein cellular localization: Lobo et al. 2009 (PMID: 19457929), He et al. 2012 (PMID: 22962422), Gil et al. 2015 (PMID: 25875300).
 - Aberrant cellular phenotypes, including defective cell migration, proliferation, and invasion: Costa et al. 2015 (PMID: 26504226)
 - Malek et al. 2017 (PMID: 29056325).

Strong

Well-established *in vitro* or *in vivo* functional studies shows no damaging effect on protein function. To be applied to intronic or synonymous variants, RNA, mini-gene or other splicing assay demonstrating no splicing impact.

Modification Disease-specific

Type:

Supporting

In vitro or *in vivo* functional study or studies showing no damaging effect on protein function.

- Phosphatase activity >0 per Mighell et al. 2018, PMID: 29706350.

Modification Disease-specific, Strength

Type:

BS4

Original ACMG Summary

Lack of segregation in affected members of a family.

Caveat: The presence of phenocopies for common phenotypes (i.e. cancer, epilepsy) can mimic lack of segregation among affected individuals. Also, families may have more than one pathogenic variant contributing to an autosomal dominant disorder, further confounding an apparent lack of segregation.

VCEP PTEN EP Specification:

Specifications: BS4: Two or more families are required for strong evidence level.

BS4_Supporting: Lack of segregation in one family.

Strong

Lack of segregation in affected members of two or more families.

Modification Disease-specific

Type:

Supporting

Lack of segregation in affected members of one family.

Modification Disease-specific, Strength

Type:

BP1

Original ACMG

Summary

Missense variant in a gene for which primarily truncating variants are known to cause disease.

Not Applicable

Comments: This rule is not applicable to PTEN.

BP2

Original ACMG

Summary

Observed in trans with a pathogenic variant for a fully penetrant dominant gene/disorder or observed in cis with a pathogenic variant in any inheritance pattern.

VCEP PTEN EP Specifications: The other variant may be either pathogenic or likely pathogenic. This rule may also be applied for at least three observations of the variant *in cis* or unknown phase with different pathogenic or likely pathogenic *PTEN* variants.

Supporting

Observed in trans with a pathogenic or likely pathogenic PTEN variant OR at least three observations in cis and/or phase unknown with different pathogenic/likely pathogenic PTEN variants.

Modification Disease-specific
Type:

BP3

Original ACMG Summary

In frame-deletions/insertions in a repetitive region without a known function.

Not Applicable

Comments: This rule is not applicable to PTEN.

BP4

Original ACMG Summary

Multiple lines of computational evidence suggest no impact on gene or gene product (conservation, evolutionary, splicing impact, etc)

Caveat: As many in silico algorithms use the same or very similar input for their predictions, each algorithm cannot be counted as an independent criterion. BP4 can be used only once in any evaluation of a variant.

VCEP **PTEN EP Specification:** To be applied to synonymous or intronic variants where SpliceAI and VarSeak *in silico* models predict no splicing impact (SpliceAI: scores 0-0.2 are considered evidence of benign. VarSeak: Class 1 and 2 are considered evidence of benign). Not to be applied for variants which may impact the intron 1 splice donor or acceptor sites, and to be used cautiously for variants which may impact the intron 6 splice acceptor. May also be applied to missense variants with REVEL score < 0.5.

PTEN EP Commentary: Per Bayesian adaptation of the ACMG/AMP variant interpretation framework (Tavtigian et al., 2018), odds of pathogenicity (OddsPath) were estimated for various numbers of previously classified controls. When REVEL scores > 0.7 were used as evidence of pathogenic and < 0.5 were used as evidence of benign, the oddsPath was equated with moderate evidence strength for benign conditions. Given that the VCEP also applies PP2 for missense variants, we decided to downgrade the evidence strength to be used at a supporting level.

Supporting

Multiple lines of computational evidence suggest no impact on gene or gene product.

- Splicing variants: Concordance of SpliceAI and VarSeak
- Missense variants: REVEL scores < 0.5

Modification Disease-specific

Type:

BP5

Original ACMG Summary

Variant found in a case with an alternate molecular basis for disease.

VCEP **PTEN EP Specifications:** At least two such cases are required for criteria to apply. In addition, the other gene/disorder must be considered highly penetrant AND the patient's personal/family history must demonstrate no overlap between the other gene and *PTEN*.

Supporting

Variant found in a case with an alternate molecular basis for disease. Other gene/disorder must be considered highly penetrant AND patient's personal/family history must demonstrate no overlap between other gene and PTEN.

Modification Disease-specific

Type:

BP6

Original ACMG Summary

Reputable source recently reports variant as benign, but the evidence is not available to the laboratory to perform an independent evaluation.

Not Applicable

This criterion is not for use as recommended by the ClinGen Sequence Variant Interpretation VCEP Review Committee.

PubMed : 29543229 [↗](#)

BP7

Original ACMG Summary

A synonymous variant for which splicing prediction algorithms predict no impact to the splice consensus sequence nor the creation of a new splice site AND the nucleotide is not highly conserved.

VCEP **PTEN EP Specification:** Intronic variants must be positioned at or

Specificationsbeyond +7/-21.

Supporting

A synonymous (silent) or intronic variant at or beyond +7/-21 for which splicing prediction algorithms predict no impact to the splice consensus sequence nor the creation of a new splice.

Modification Disease-specific
Type:

Rules for Combining Criteria

Pathogenic

- 1 Very Strong** (PVS1, PS2_Very Strong, PS4_Very Strong, PM6_Very Strong) **AND** **≥ 1 Strong** (PVS1_Strong, PS1, PS2, PS3, PS4, PM6_Strong, PP1_Strong)
- 1 Very Strong** (PVS1, PS2_Very Strong, PS4_Very Strong, PM6_Very Strong) **AND** **≥ 2 Moderate** (PVS1_Moderate, PS3_Moderate, PS4_Moderate, PM1, PM4, PM5, PM6, PP1_Moderate)
- 1 Very Strong** (PVS1, PS2_Very Strong, PS4_Very Strong, PM6_Very Strong) **AND 1 Moderate** (PVS1_Moderate, PS3_Moderate, PS4_Moderate, PM1, PM4, PM5, PM6, PP1_Moderate) **AND 1 Supporting** (PS3_Supporting, PS4_Supporting, PM2_Supporting, PP1, PP2, PP3)
- 1 Very Strong** (PVS1, PS2_Very Strong, PS4_Very Strong, PM6_Very Strong) **AND** **≥ 2 Supporting** (PS3_Supporting, PS4_Supporting, PM2_Supporting, PP1, PP2, PP3)
- ≥ 2 Strong** (PVS1_Strong, PS1, PS2, PS3, PS4, PM6_Strong, PP1_Strong)
- 1 Strong** (PVS1_Strong, PS1, PS2, PS3, PS4, PM6_Strong, PP1_Strong) **AND** **≥ 3 Moderate** (PVS1_Moderate, PS3_Moderate, PS4_Moderate, PM1, PM4, PM5, PM6, PP1_Moderate)
- 1 Strong** (PVS1_Strong, PS1, PS2, PS3, PS4, PM6_Strong, PP1_Strong) **AND 2 Moderate** (PVS1_Moderate, PS3_Moderate, PS4_Moderate, PM1, PM4, PM5, PM6, PP1_Moderate) **AND** **≥ 2 Supporting** (PS3_Supporting, PS4_Supporting, PM2_Supporting, PP1, PP2, PP3)
- 1 Strong** (PVS1_Strong, PS1, PS2, PS3, PS4, PM6_Strong, PP1_Strong) **AND 1 Moderate** (PVS1_Moderate, PS3_Moderate, PS4_Moderate, PM1, PM4, PM5, PM6, PP1_Moderate) **AND** **≥ 4 Supporting** (PS3_Supporting, PS4_Supporting, PM2_Supporting, PP1, PP2, PP3)

Likely Pathogenic

- 1 Very Strong** (PVS1, PS2_Very Strong, PS4_Very Strong, PM6_Very Strong) **AND 1 Moderate** (PVS1_Moderate, PS3_Moderate, PS4_Moderate, PM1, PM4, PM5, PM6, PP1_Moderate)
- 1 Strong** (PVS1_Strong, PS1, PS2, PS3, PS4, PM6_Strong, PP1_Strong) **AND 1 Moderate** (PVS1_Moderate, PS3_Moderate, PS4_Moderate, PM1, PM4, PM5, PM6, PP1_Moderate)
- 1 Strong** (PVS1_Strong, PS1, PS2, PS3, PS4, PM6_Strong, PP1_Strong) **AND** **≥ 2 Supporting** (PS3_Supporting, PS4_Supporting, PM2_Supporting, PP1, PP2, PP3)
- ≥ 3 Moderate** (PVS1_Moderate, PS3_Moderate, PS4_Moderate, PM1, PM4, PM5, PM6, PP1_Moderate)
- 2 Moderate** (PVS1_Moderate, PS3_Moderate, PS4_Moderate, PM1, PM4, PM5, PM6, PP1_Moderate) **AND** **≥ 2 Supporting** (PS3_Supporting, PS4_Supporting, PM2_Supporting, PP1, PP2, PP3)
- 1 Moderate** (PVS1_Moderate, PS3_Moderate, PS4_Moderate, PM1, PM4, PM5, PM6, PP1_Moderate) **AND** **≥ 4 Supporting** (PS3_Supporting, PS4_Supporting, PM2_Supporting, PP1, PP2, PP3)
- 1 Strong** (PVS1_Strong, PS1, PS2, PS3, PS4, PM6_Strong, PP1_Strong) **AND 2 Moderate** (PVS1_Moderate, PS3_Moderate, PS4_Moderate, PM1, PM4, PM5, PM6, PP1_Moderate)
- 1 Very Strong** (PVS1, PS2_Very Strong, PS4_Very Strong, PM6_Very Strong) **AND 1 Supporting**

1 Very Strong (*PS1_Very Strong, PS2_Very Strong, PS4_Very Strong, PM6_Very Strong*) **AND 1 Supporting**
(*PS3_Supporting, PS4_Supporting, PM2_Supporting, PP1, PP2, PP3*)

Benign

≥ 2 Strong (*BS1, BS2, BS3, BS4*)

1 Stand Alone (*BA1*)

Likely Benign

≥ 2 Supporting (*BS1_Supporting, BS2_Supporting, BS3_Supporting, BS4_Supporting, BP2, BP4, BP5, BP7*)

1 Strong (*BS1, BS2, BS3, BS4*)

Files & Images

PVS1_DecisionTree: [!\[\]\(05be7c7a8995decd503647c99211f7c2_img.jpg\)](#)

RULES FOR COMBINING CRITERIA FOR CLASSIFICATION:

RULES FOR COMBINING CRITERIA FOR CLASSIFICATION

Variants will be classified per Richards et al., 2015 with the following exceptions:

- 1 Benign Strong (BS) = Likely Benign.
- 1 Pathogenic Very Strong (PVS) + 1 Pathogenic Supporting (PP) = Likely Pathogenic.
- Variants with conflicting evidence may be classified using the Bayesian points system (Tavtigian et al., 2018).

PTEN Phenotype Scoring for Pediatric Patients:

TABLE 2 PTEN phenotype scoring for pediatric patients

Feature	Score (points)
Macrocephaly of >2 SD to <4 SD	2
Extreme macrocephaly (≥ 4 SD)	3
PTEN-specific MRI characteristics (dilated Virchow-Robin, prominent perivascular spaces)	2
Autism/developmental delay (DD)/intellectual disability (ID)	2
Penile freckling	3
Lipoma	1
Oral papilloma	3
Hamartomatous polyp(s)	3
Arteriovenous malformation/hemangioma	2
Thyroid cancer	3
Thyroid nodule/Hashimoto's thyroiditis	2

Cleveland Clinic score:

Feature	Score	Feature	Score
Neurological		Breast	
Macrocephaly	6	Cancer dx <40 yrs	4
Extreme macrocephaly (men: ≥ 63 cm, women: ≥ 60 cm)	10	Cancer dx 40-49 yrs	2
Lhermitte-Duclos disease	10	Cancer dx ≥ 50 yrs	1
Autism/developmental delay	1	Fibrocystic breast disease	1
Skin		Thyroid	
Trichilemmoma (biopsy-proven)	10	Cancer dx <20 yrs	10
Oral papillomas	6	Cancer dx 20-49 yrs	4
Penile freckling	6	Cancer dx ≥ 50 yrs	1
Acral keratoses	1	Goiter, nodules, or Hashimoto's thyroiditis	4
Lipoma	1	Genitourinary	
Arteriovenous malformation	6	Endometrial cancer dx 20-29 yrs	10
Gastrointestinal		Endometrial cancer dx 30-49 yrs	6
≥ 5 gastrointestinal polyps, any type	6	Endometrial cancer dx ≥ 50 yrs	1
Hamartoma or ganglioneuroma	10	Uterine fibroids	1
Glycogenic acanthosis	10	Renal cell carcinoma	1

BLOSUM matrix:

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13. Stambolic V Suzuki A et al. *Negative regulation of PKB/Akt-dependent cell survival by the tumor suppressor PTEN.* **Cell** (1998) 95 (1) p. 29-39. 10.1016/s0092-8674(00)81780-8 9778245 [↗](#)
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